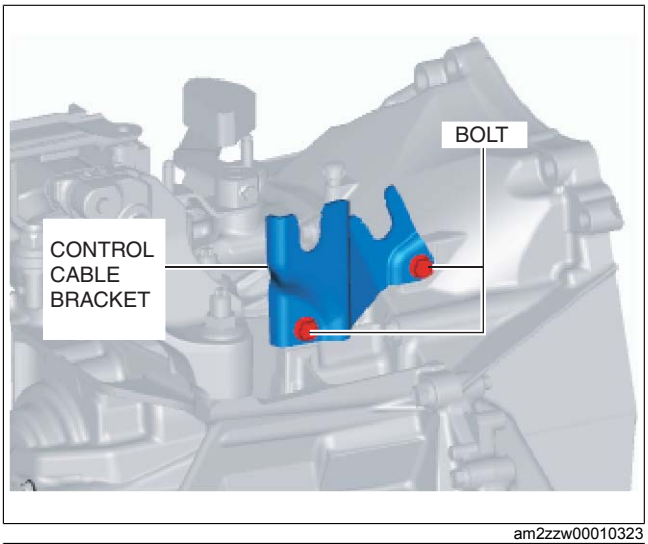
MANUAL TRANSAXLE REMOVAL/INSTALLATION [F66M-R (SKYACTIV-D 1.5)]

id0515001600n4

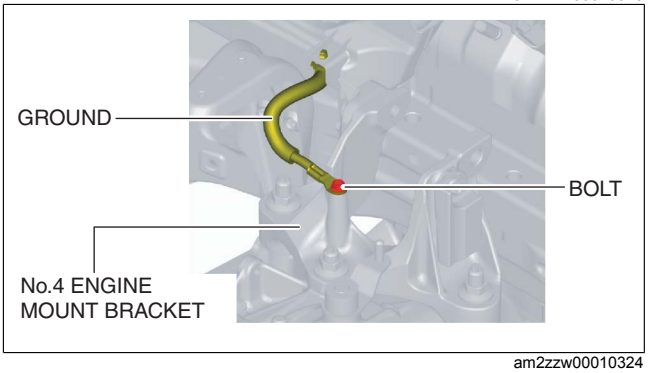
Removal

1. Disconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/ CONNECTION.)
2. Remove the engine cover. (See ENGINE COVER REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
3. Remove the front under cover No.2. (See FRONT UNDER COVER No.2 REMOVAL/INSTALLATION.)
4. Remove the front under cover No.1. (See FRONT UNDER COVER No.1 REMOVAL/INSTALLATION.)
5. Drain the manual transaxle oil. (See MANUAL TRANSAXLE OIL REPLACEMENT [F66M-R].)
6. Disconnect and/or remove the following parts in the engine compartment.
   (1) Remove the air cleaner, air hose and fresh air duct as a single unit. (See INTAKE-AIR SYSTEM REMOVAL/ INSTALLATION [SKYACTIV-D 1.5].)
   (2) Remove the battery and battery tray. (See BATTERY REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
   (3) Remove the DC-DC converter. (with i-ELOOP) (See DC-DC CONVERTER (i-ELOOP) REMOVAL/ INSTALLATION [WITH i-ELOOP (SKYACTIV-D 1.5)].)
   (4) Disconnect the control cable from the manual transaxle. (See MANUAL TRANSAXLE SHIFT MECHANISM REMOVAL/INSTALLATION [F66M-R].)
   (5) Remove the control cable bracket from the manual transaxle.
   (6) Remove the turbocharger air outlet pipe. (See INTAKE-AIR SYSTEM REMOVAL/ INSTALLATION [SKYACTIV-D 1.5].)



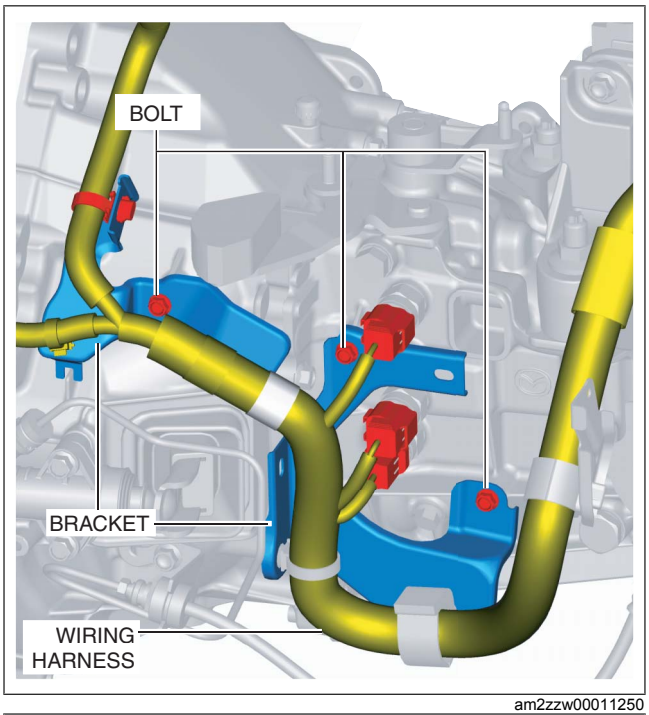
am2zzw00010323

   (7) Remove the ground from the No.4 engine mount bracket.
   (8) Disconnect the neutral switch No.1 connector from manual transaxle. (See NEUTRAL SWITCH REMOVAL/INSTALLATION [F66M-R].)
   (9) Disconnect the neutral switch No.2 connector from manual transaxle. (See NEUTRAL SWITCH REMOVAL/INSTALLATION [F66M-R].)
   (10) Disconnect the back-up light switch connector from the manual transaxle. (See BACK-UP LIGHT SWITCH REMOVAL/INSTALLATION [F66M-R].)



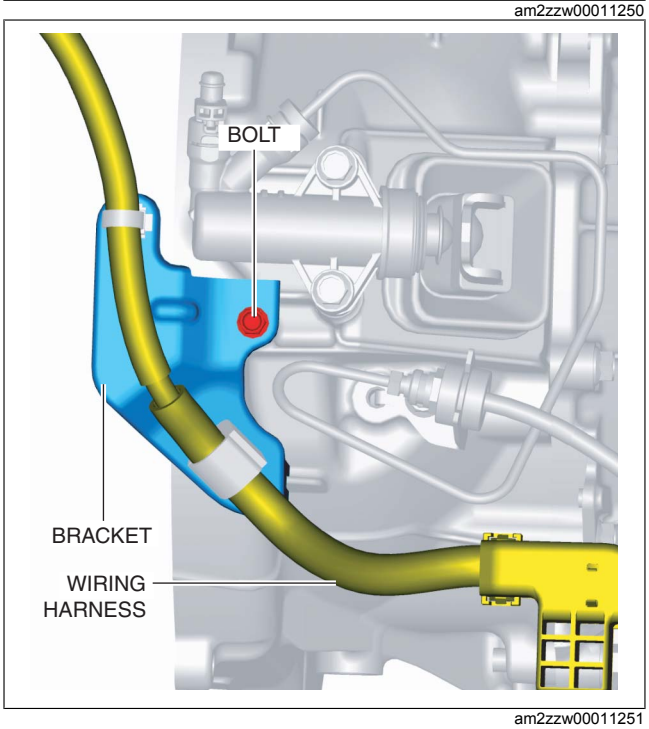
am2zzw00010324

(11) Remove the brackets shown in the figure from the manual transaxle and set the wiring harness aside.



am2zzw00011250

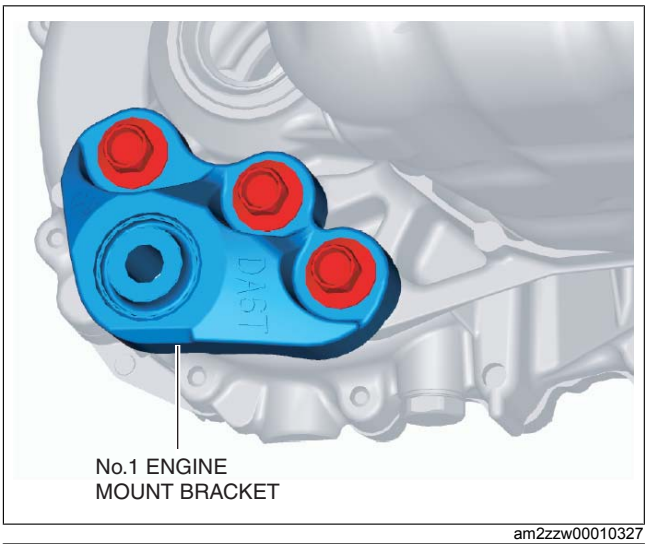
(12) Remove the bracket shown in the figure from the manual transaxle and set the wiring harness aside. (with i-ELOOP)



am2zzw00011251

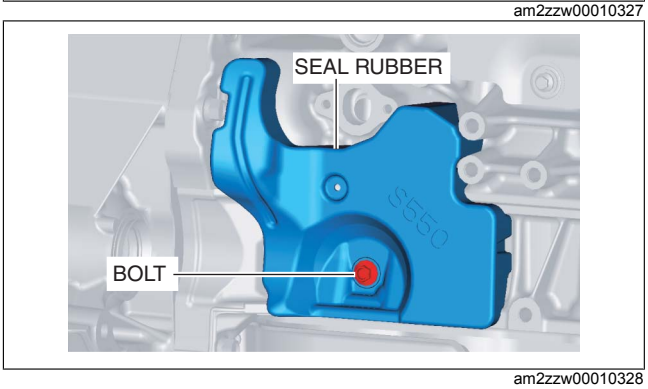
7. Disconnect and/or remove the following parts related to the suspension and axle.
   (1) Remove the front tires. (See GENERAL PROCEDURES (FRONT AND REAR AXLES).)
   (2) Remove the splash shield. (See SPLASH SHIELD REMOVAL/INSTALLATION.)
   (3) Remove the clip used to secure the brakehose from the front shock absorber and set the brake hose aside. (See BRAKE HOSE (FRONT) REMOVAL/INSTALLATION.)
   (4) Disconnect the ABS wheel-speed sensors from the steering knuckles. (See FRONT ABS WHEEL-SPEED SENSOR REMOVAL/INSTALLATION.)
   (5) Disconnect the tie-rod end ball joints from the steering knuckles. (See FRONT CROSSMEMBER REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
   (6) Disconnect the front stabilizer control link from the front shock absorber. (See FRONT STABILIZER REMOVAL.)
   (7) Disconnect the front lower arms from the steering knuckles. (See FRONT CROSSMEMBER REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
   (8) Disconnect the drive shaft (LH) from the manual transaxle. (See FRONT DRIVE SHAFT REMOVAL/INSTALLATION.)
   (9) Disconnect the drive shaft (RH) from the manual transaxle. (See FRONT DRIVE SHAFT REMOVAL/INSTALLATION.)
8. Disconnect and/or remove the following parts from the underside of the vehicle.
   (1) Remove the clutch release cylinder with the clutch pipe still connected and set it out of the way. (See CLUTCH RELEASE CYLINDER REMOVAL/INSTALLATION [F66M-R].)
   (2) Remove the starter. (See STARTER REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
   (3) Remove the front crossmember component and No.1 engine mount rubber as a single unit. (See FRONT CROSSMEMBER REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)

(4) Remove the No.1 engine mount bracket from the manual transaxle.



am2zzw00010327

(5) Remove the seal rubber.



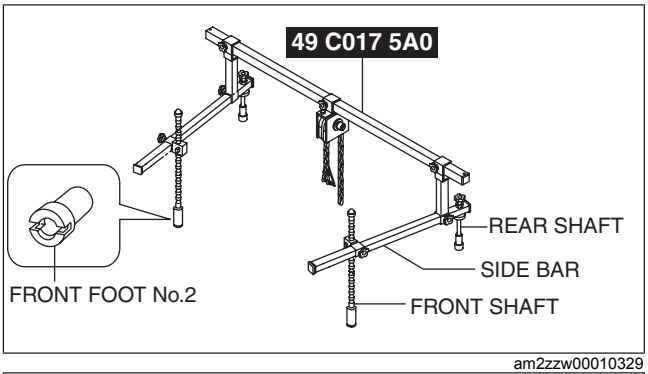
am2zzw00010328

9. Install the **SST (49 C017 5A0, 49 UN30 3050, 49 L017 5A0)** using the following procedure.

Caution

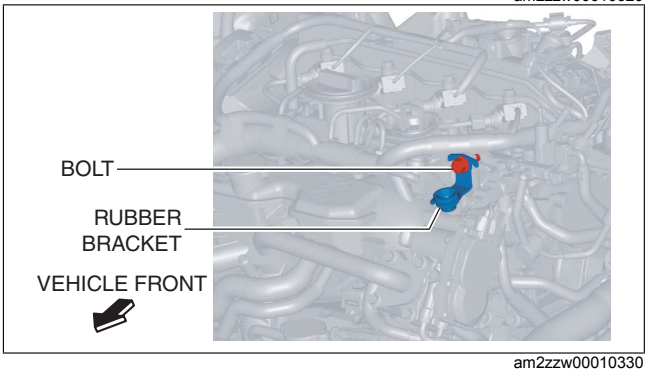
- **Refer to the SST (49 C017 5A0) instruction manual for the basic handing procedure.**

(1) Install one front foot No.2 to each of the left and right front shafts of the **SST (49 C017 5A0)**.



am2zzw00010329

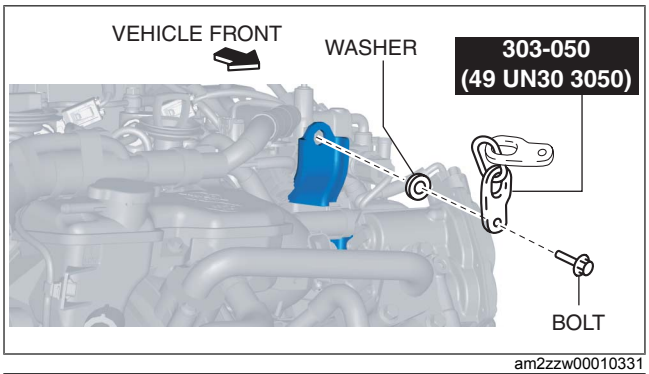
(2) Remove the rubber bracket.



am2zzw00010330

(3) Install the **SST (49 UN30 3050)** to the position shown in the figure using the following bolt and washer.

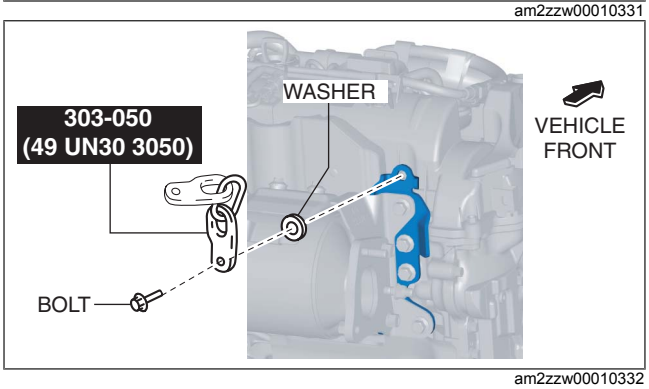
Bolt
Part number 99794 1025 or an M10 × 1.25, length 25 mm {0.98 in}
Washer
approx. 3 mm {0.1 in} thickness
Tightening torque
38—52 N·m {3.9—5.3 kgf·m, 29—38 ft·lbf}



am2zzw00010331

(4) Install the **SST (49 UN30 3050)** to the position shown in the figure using the following bolt and washer.

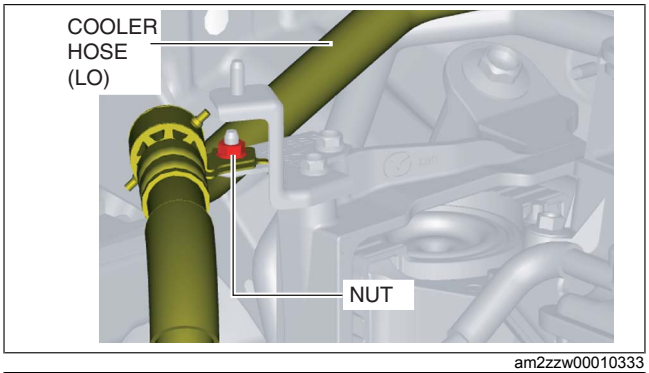
Bolt
Part number 99794 1025 or an M10 × 1.25, length 25 mm {0.98 in}
Washer
approx. 3 mm {0.1 in} thickness
Tightening torque
38—52 N·m {3.9—5.3 kgf·m, 29—38 ft·lbf}



am2zzw00010332

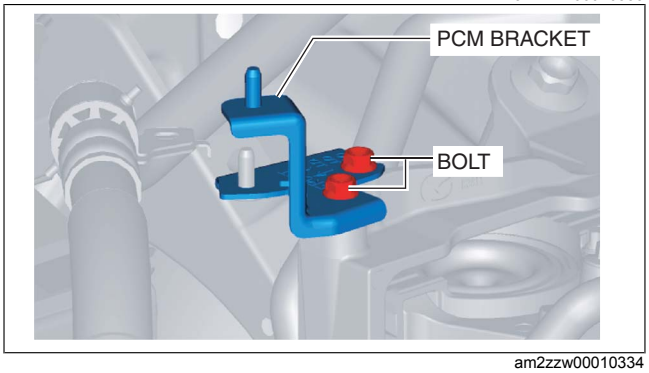
(5) Remove the PCM component with the PCM connector remaining connected, and set the PCM component aside. (See PCM REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)

(6) Remove the nut shown in the figure and set the cooler hose (LO) aside.



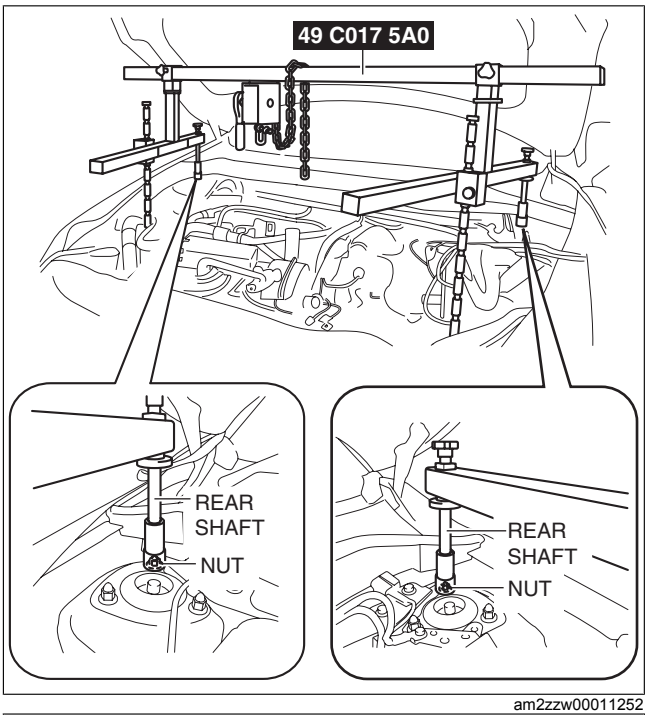
am2zzw00010333

(7) Remove the PCM bracket.



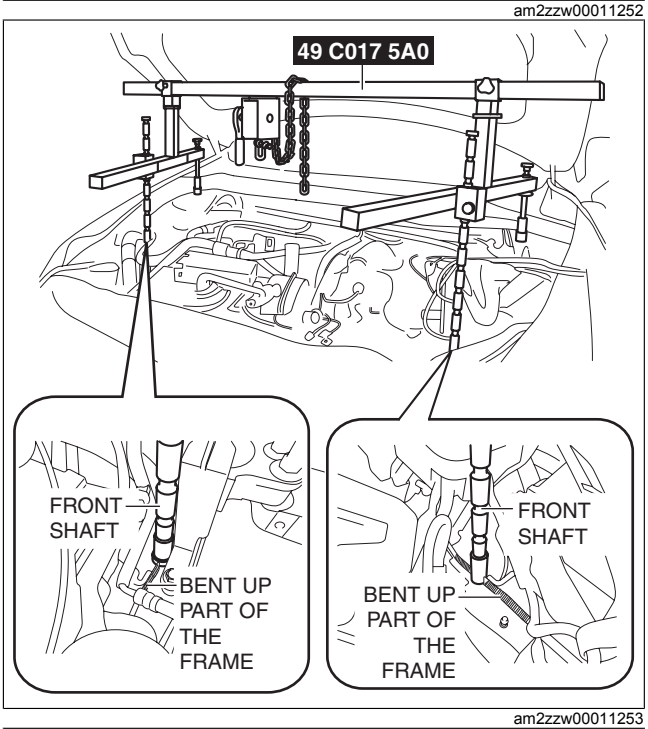
am2zzw00010334

(8) Set the rear shafts of the **SST (49 C017 5A0)** on the left/right front shock absorber nuts (top) as shown in the figure.



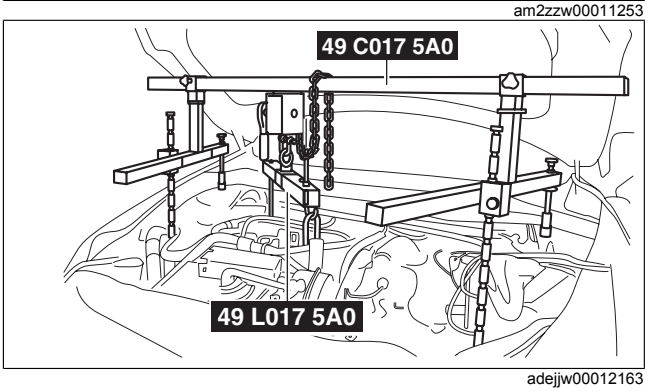
am2zzw00011252

(9) Set the front shafts of the **SST (49 C017 5A0)** on the bent up part of the frame as shown in the figure.



am2zzw00011253

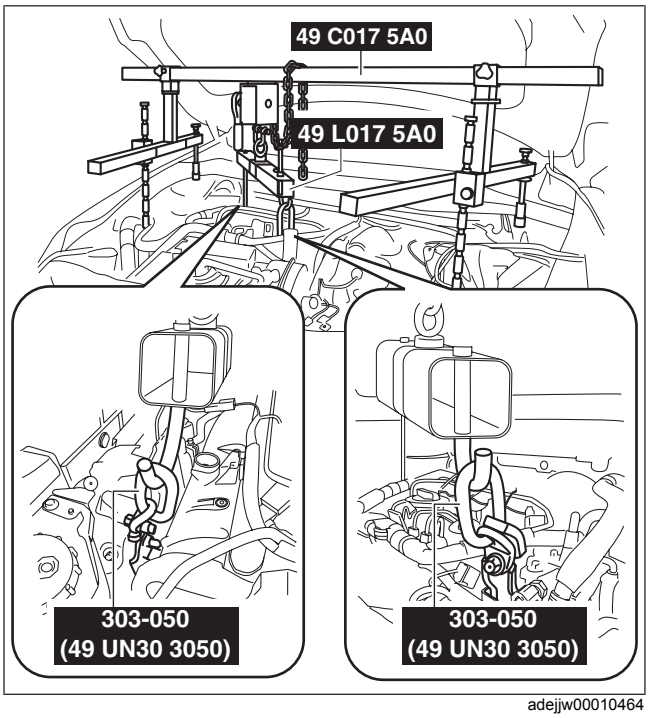
(10) Install the **SST (49 L017 5A0)** to the **SST (49 C017 5A0)** as shown in the figure.



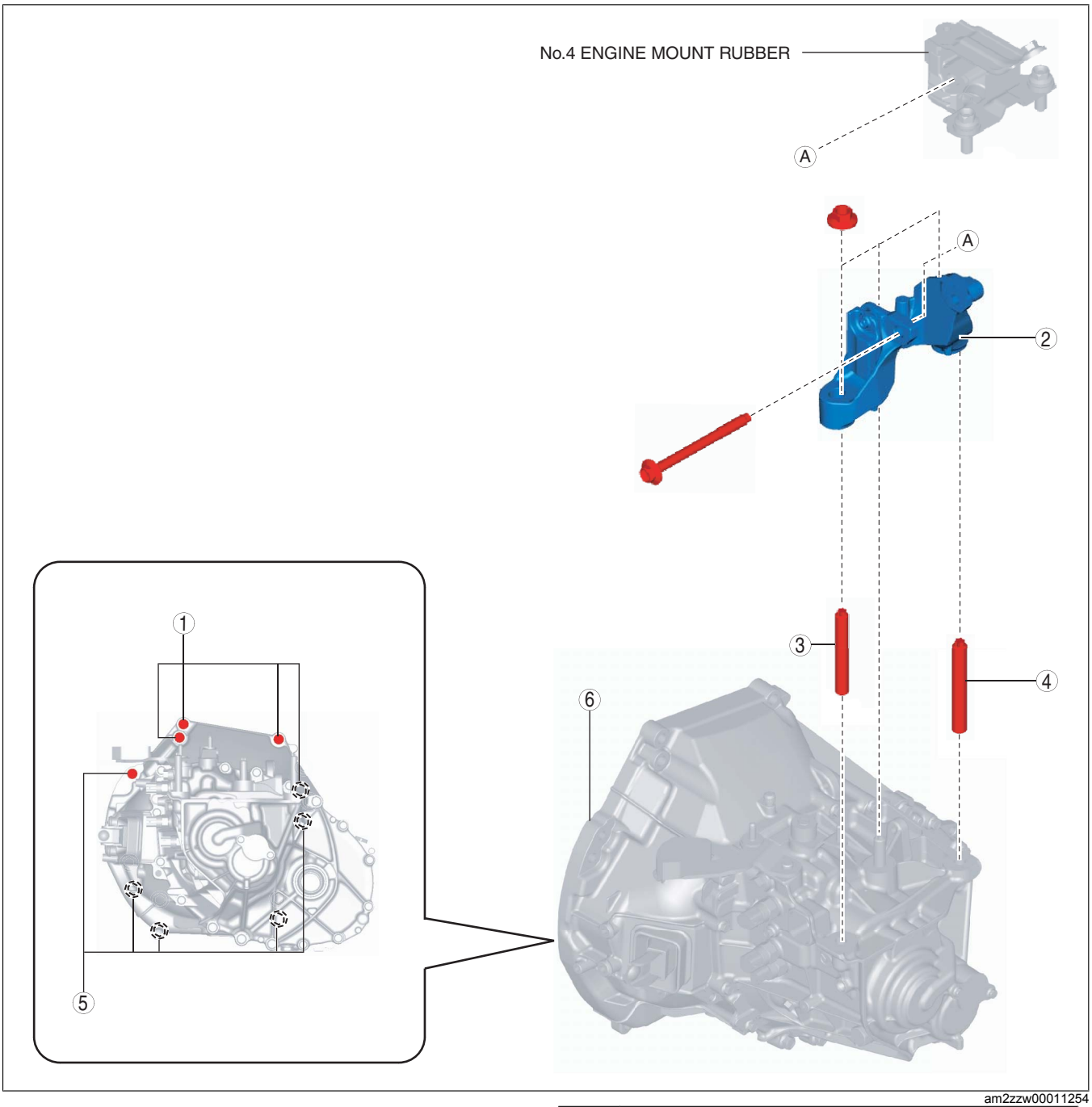
adejjw00012163

(11) Install the **SST (49 L017 5A0)** to the **SST (49 UN30 3050)** with the hook of the **SST (49 L017 5A0)** facing outward.
(12) Adjust the height of the left and right side bars so that they are leveled, then tighten each part of the **SST (49 C017 5A0).**
(13) Apply tension to the chain to secure the engine.

10. Remove in the order indicated in the table.



adejjw00010464



am2zzw00011254

1	Transaxle mounting bolt (upper side)
2	No.4 engine mount bracket (See No.4 engine mount bracket removal note.)
3	Stud bolt No.1
4	Stud bolt No.3

5	Transaxle mounting bolt (lower side) (See Transaxle mounting bolt removal note.)
6	Manual transaxle

No.4 engine mount bracket removal note

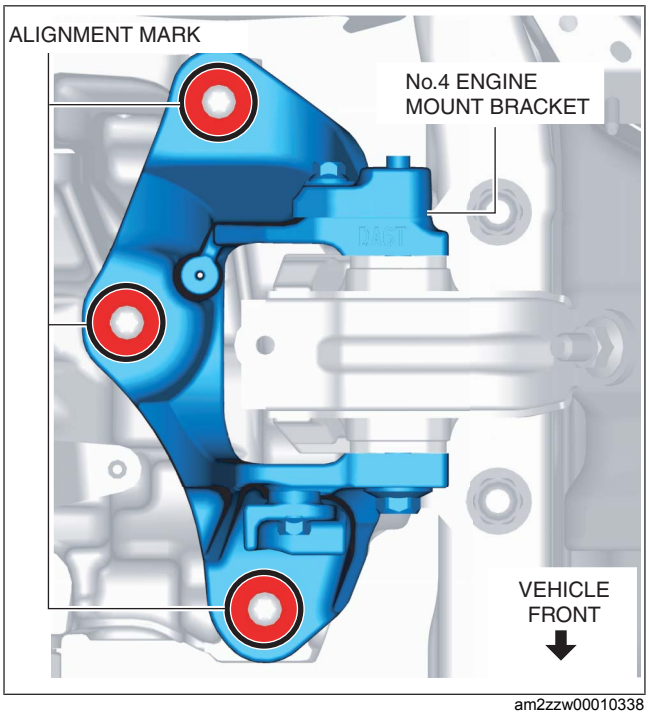
Caution

- **A slot has been adopted for the No.4 engine mount bracket installation hole. If the No.4 engine mount bracket is deviated from the original position when installing the No.4 engine mount, engine noise or vibration could increase. When removing the No.4 engine mount, place alignment marks on the No.4 engine mount bracket so that they can be assembled to the same positions as before removal.**

1. Place alignment marks on the locations shown in the figure so that they can be assembled to the same positions as before removal.

   Note
   - Paint so that the outline of the nut is framed on the bracket side.

2. Remove the No.4 engine mount bracket.

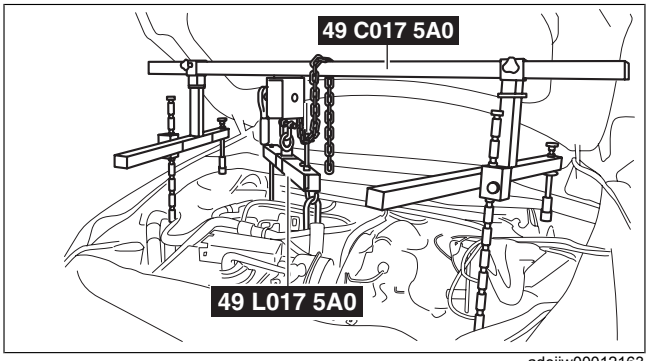


am2zzw00010338

Transaxle mounting bolt removal note

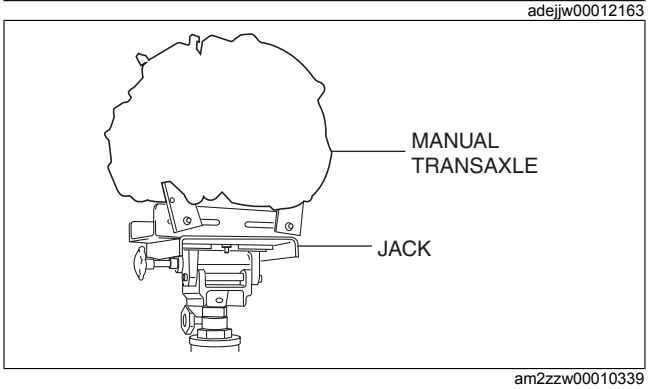
Warning
- **Remove the manual transaxle carefully, holding it steady. If the manual transaxle falls it could be damaged or cause injury.**

1. Adjust the **SST (49 C017 5A0)** and lean the engine toward the manual transaxle.



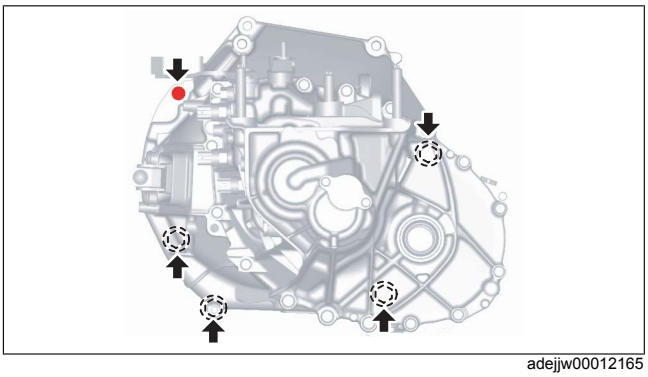
adejjw00012163

2. Support the manual transaxle on a jack.



am2zzw00010339

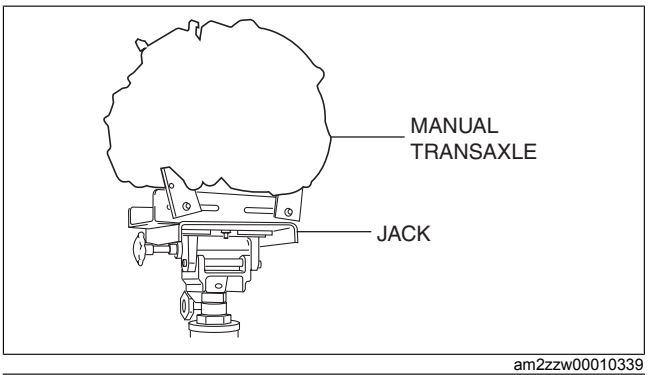
3. Remove the transaxle mounting bolts (lower side).
4. Remove the manual transaxle.


adejjw00012165

Installation

Warning
- **Remove the manual transaxle carefully, holding it steady. If the manual transaxle falls it could be damaged or cause injury.**

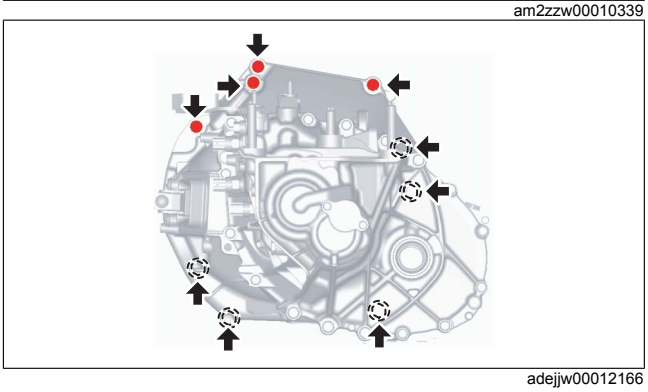
1. Set the manual transaxle on a jack and lift into place.



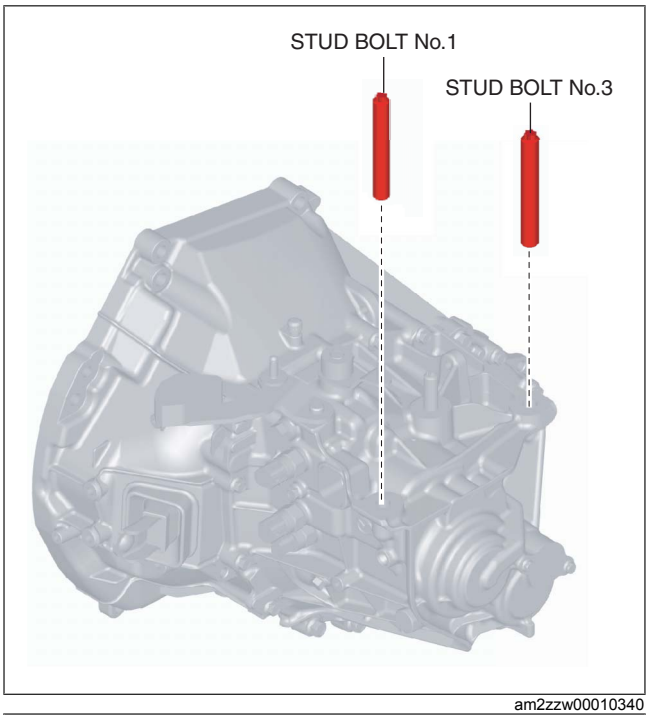
am2zzw00010339

2. Install the manual transaxle to the engine, and tighten the transaxle mounting bolts.

   Tightening torque
   38—52 N·m {3.9—5.3 kgf·m, 29—38 ft·lbf}


adejjw00012166

3. Install stud bolts No.1 and No.3.

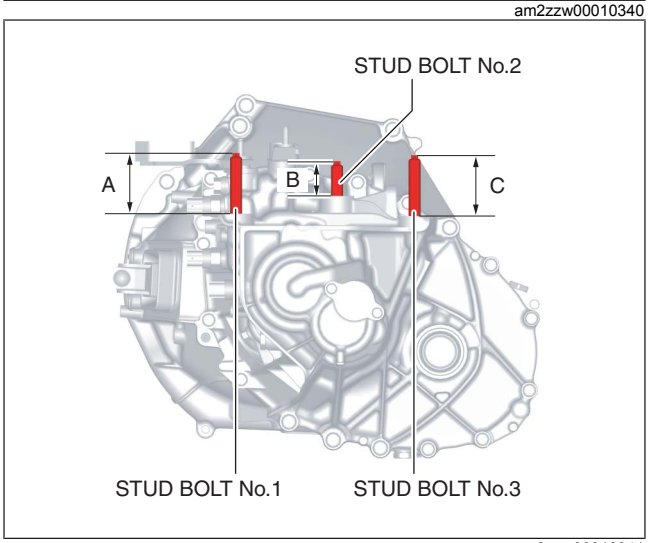


am2zzw00010340

4. Measure the projection of the stud bolts.

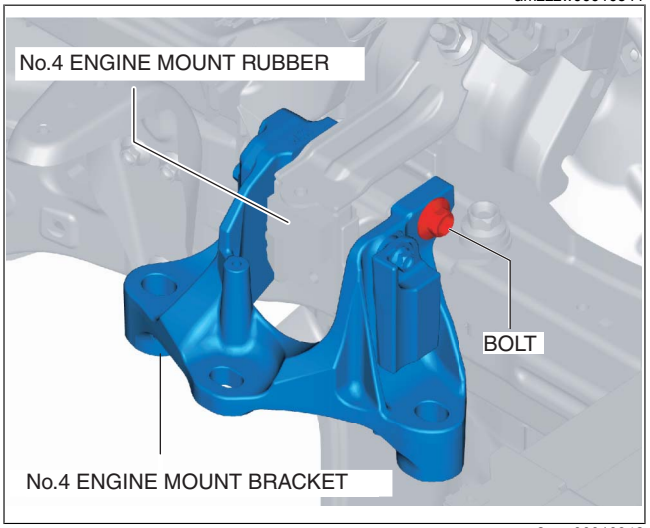
Projection A of stud bolt No.1
70.7—72.7 mm {2.79—2.86 in}
Projection B of stud bolt No.2
40.0—42.0 mm {1.58—1.65 in}
Projection C of stud bolt No.3
70.7—72.7 mm {2.79—2.86 in}

- If the projection amount is not within the range, adjust the projection amount of the stud bolt.



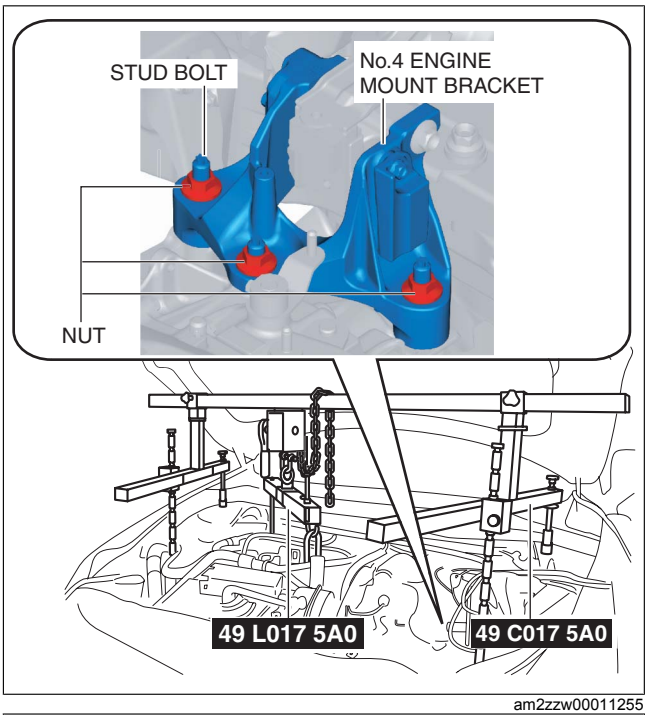
am2zzw00010341

5. Install the No.4 engine mount bracket to No.4 engine mount rubber, and temporarily tighten the installation bolt.



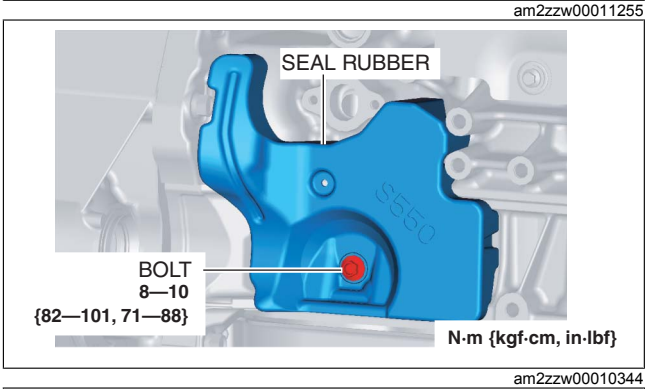
am2zzw00010342

6. Lift up the manual transaxle using the **SST (49 C017 5A0)**, pass the stud bolt through the No.4 engine mount bracket, and temporarily tighten the No.4 engine mount bracket installation nuts.



am2zzw00011255

7. Install the seal rubber.

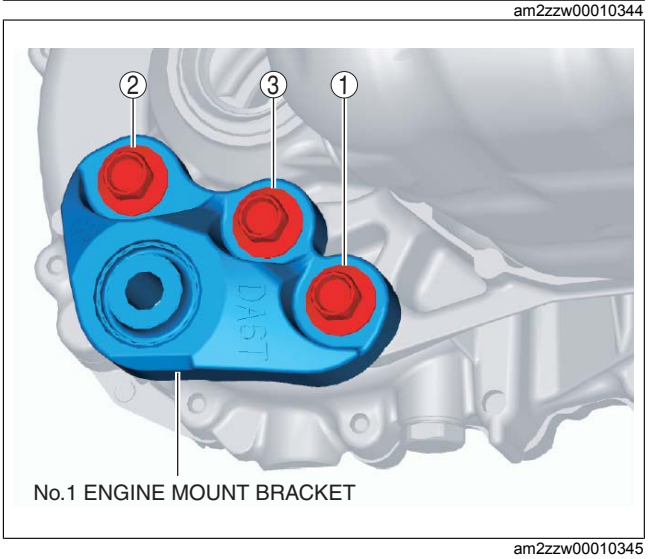


am2zzw00010344

8. Install the No.1 engine mount bracket to the manual transaxle using the procedure shown in the figure, and tighten the bolts.

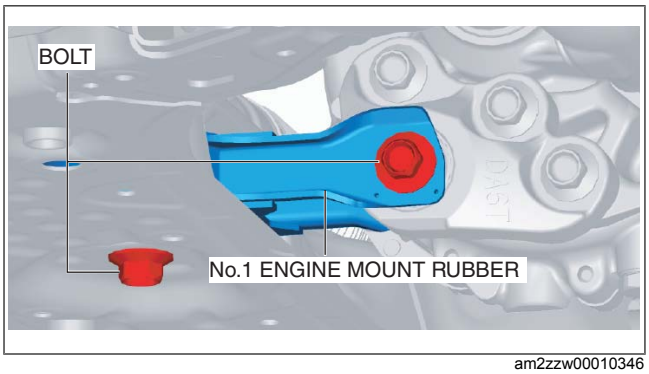
   Tightening torque
   84—98 N·m {8.6—9.9 kgf·m, 62—72 ft·lbf}

9. Install the front crossmember component and No.1 engine mount rubber as a single unit. (See FRONT CROSSMEMBER REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)



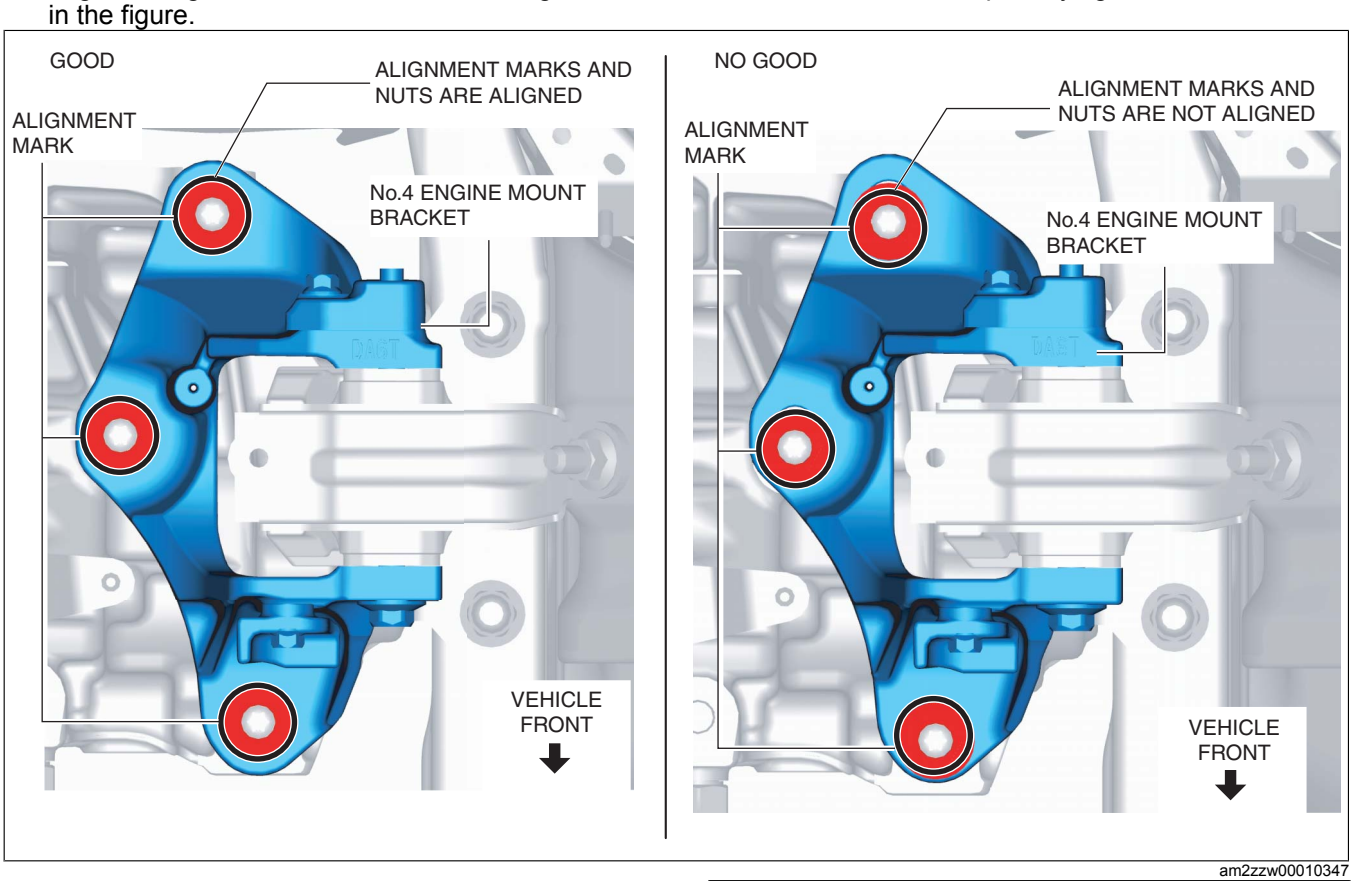
am2zzw00010345

10. Temporarily tighten the No.1 engine mount rubber installation bolts.



am2zzw00010346

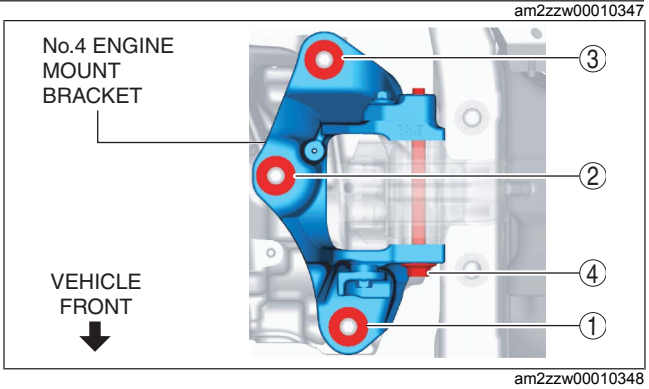
11. Align the alignment marks on the No.4 engine mount bracket and nuts, and temporarily tighten the nuts shown in the figure.



am2zzw00010347

12. Tighten the No.4 engine mount bracket installation nuts and bolt in the order shown in the figure.

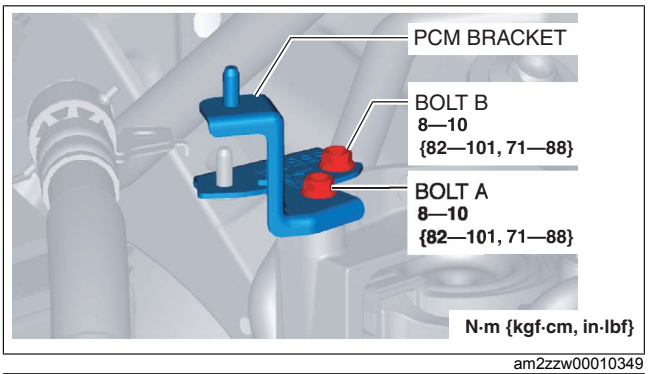
	Tightening torque
1, 2, 3	92—116 N·m {9.4—11 kgf·m, 68—85 ft·lbf}
4	81—99 N·m {8.3—10 kgf·m, 60—73 ft·lbf}



am2zzw00010348

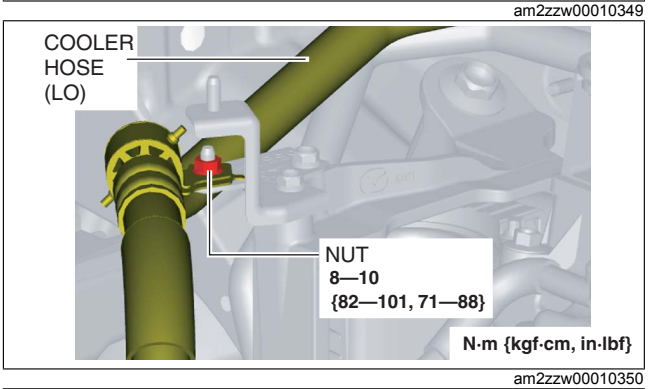
13. Remove the **SST (49 C017 5A0, 49 L017 5A0)**.

14. Install the PCM bracket in the following order.
    (1) Tighten bolt A and position the PCM bracket.
    (2) Tighten in the order of bolts B, A.



am2zzw00010349

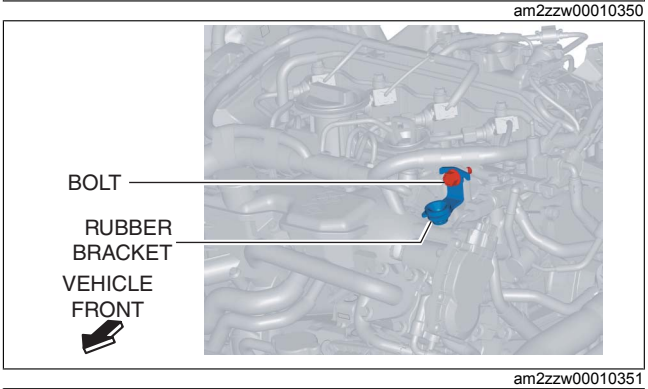
15. Install the cooler hose (LO).
16. Install the PCM component. (See PCM REMOVAL/ INSTALLATION [SKYACTIV-D 1.5].)



am2zzw00010350

17. Install the rubber bracket.

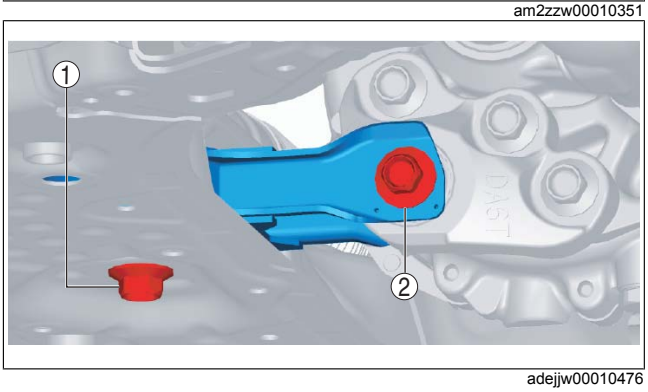
    Tightening torque
    37—52 N·m {3.8—5.3 kgf·m, 28—38 ft·lbf}



am2zzw00010351

18. Tighten the No.1 engine mount rubber installation bolts in the order shown in the figure.

	Tightening torque
1	141—172 N·m {15—17 kgf·m, 104—126 ft·lbf}
2	130—164 N·m {14—16 kgf·m, 96—120 ft·lbf}

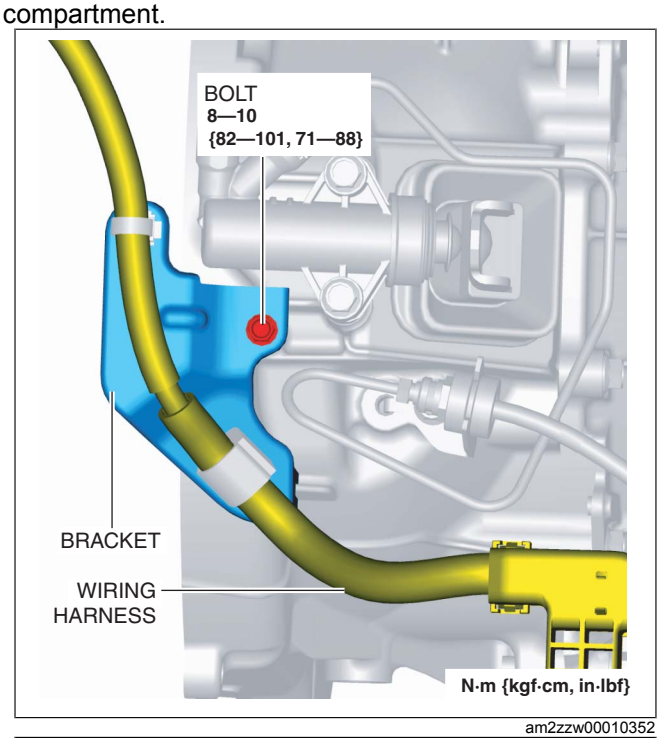


adejjw00010476

19. Connect and/or install the following parts to the underside of the vehicle.
    (1) Install the starter. (See STARTER REMOVAL/ INSTALLATION [SKYACTIV-D 1.5].)
    (2) Install the clutch release cylinder to the manual transaxle. (See CLUTCH RELEASE CYLINDER REMOVAL/ INSTALLATION [F66M-R].)
20. Connect and/or install the following parts related to the suspension and axle.
    (1) Install the drive shaft (RH) to the manual transaxle. (See FRONT DRIVE SHAFT REMOVAL/ INSTALLATION.)
    (2) Install the drive shaft (LH) to the manual transaxle. (See FRONT DRIVE SHAFT REMOVAL/ INSTALLATION.)
    (3) Install the front lower arms to the steering knuckles. (See FRONT CROSSMEMBER REMOVAL/ INSTALLATION [SKYACTIV-D 1.5].)
    (4) Install the front stabilizer control link to the front shock absorber. (See FRONT STABILIZER INSTALLATION.)
    (5) Install the tie-rod end ball joints to the steering knuckles. (See FRONT CROSSMEMBER REMOVAL/ INSTALLATION [SKYACTIV-D 1.5].)

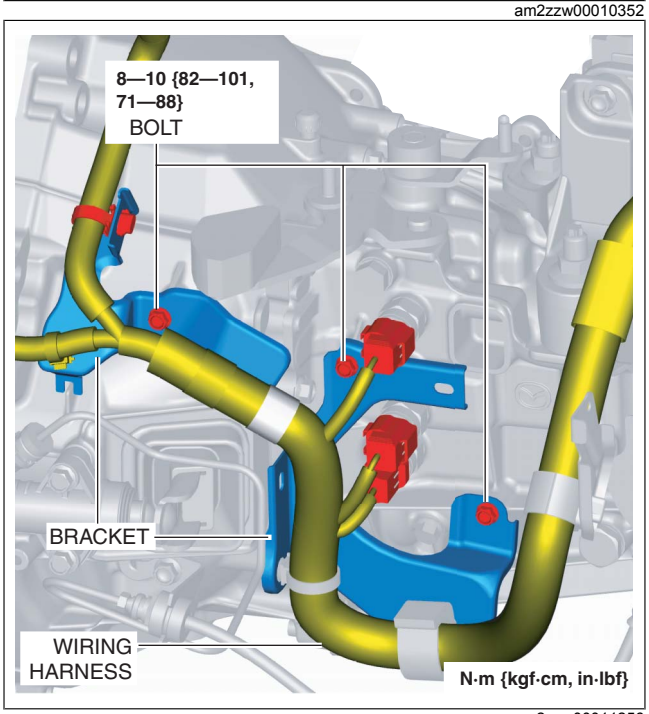
(6) Install the ABS wheel-speed sensors to the steering knuckles. (See FRONT ABS WHEEL-SPEED SENSOR REMOVAL/INSTALLATION.)
(7) Install the brake hose to the front shock absorber. (See BRAKE HOSE (FRONT) REMOVAL/INSTALLATION.)
(8) Install the splash shield. (See SPLASH SHIELD REMOVAL/INSTALLATION.)
(9) Install the front tires. (See GENERAL PROCEDURES (FRONT AND REAR AXLES).)

21. Connect and/or install the following parts in the engine compartment.
(1) Install the bracket shown in the figure to the manual transaxle. (with i-ELOOP)



am2zzw00010352

(2) Install the brackets to the manual transaxle.
(3) Connect the neutral switch No.2 connector. (See NEUTRAL SWITCH REMOVAL/INSTALLATION [F66M-R].)
(4) Connect the neutral switch No.1 connector. (See NEUTRAL SWITCH REMOVAL/INSTALLATION [F66M-R].)
(5) Connect the back-up light switch connector. (See BACK-UP LIGHT SWITCH REMOVAL/INSTALLATION [F66M-R].)

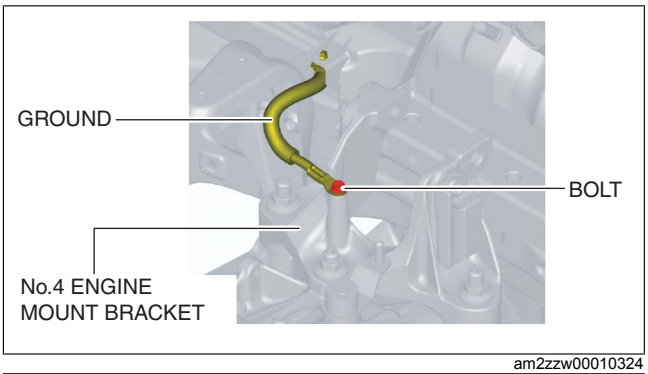


am2zzw00011256

(6) Install the ground to the No.4 engine mount bracket.

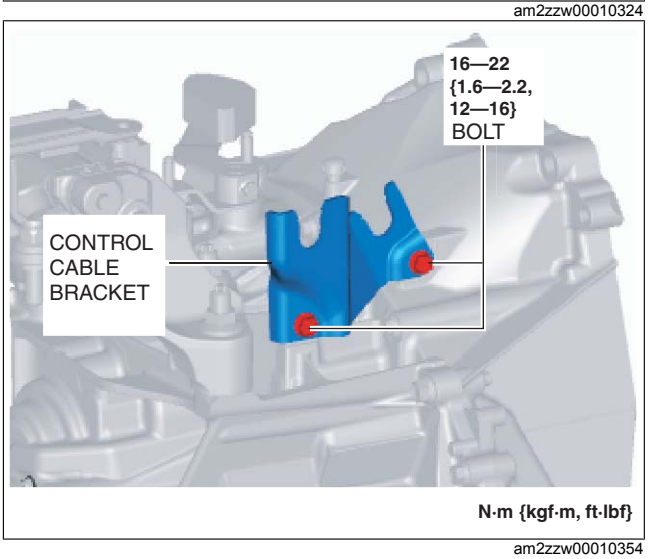
Tightening torque
9—12 N·m {92—122 kgf·cm, 80—106 in·lbf}

(7) Install the turbocharger air outlet pipe. (See INTAKE-AIR SYSTEM REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)



am2zzw00010324

(8) Install the control cable bracket to the manual transaxle.
(9) Install the control cable to the manual transaxle. (See MANUAL TRANSAXLE SHIFT MECHANISM REMOVAL/INSTALLATION [F66M-R].)
(10) Install the DC-DC converter. (with i-ELOOP) (See DC-DC CONVERTER (i-ELOOP) REMOVAL/INSTALLATION [WITH i-ELOOP (SKYACTIV-D 1.5)].)
(11) Install the battery and battery tray. (See BATTERY REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
(12) Install the air cleaner, air hose and fresh air duct as a single unit. (See INTAKE-AIR SYSTEM REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)



am2zzw00010354

22. Install the front under cover No.1. (See FRONT UNDER COVER No.1 REMOVAL/INSTALLATION.)
23. Add the specified amount of manual transaxle oil. (See MANUAL TRANSAXLE OIL REPLACEMENT [F66M-R].)
24. Install the front under cover No.2. (See FRONT UNDER COVER No.2 REMOVAL/INSTALLATION.)
25. Connect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.)
26. Install the engine cover. (See ENGINE COVER REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
27. If the manual transaxle is overhauled, perform the “INSPECTION AFTER MANUAL TRANSAXLE OVERHAUL”. (See INSPECTION AFTER MANUAL TRANSAXLE OVERHAUL [F66M-R].)